Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Practice Worksheet

Chapter 1.3 (Storage & Compression) & Chapter 2.2, 2.3 (Transmission & Security)

[TEACHER ANSWER KEY]

Date: _____ **Student Name:** TEACHER ANSWER KEY & MARK SCHEME

Class: Secondary 2 (Sec 2) **Total Marks: 40 Marks | Obtained: 40 / 40**

SECTION 1: DATA STORAGE & FILE SIZE CALCULATIONS [10 MARKS]

1. Unit Conversions: (3 Marks)

Convert the following storage capacities into the requested units:

a) Convert 4 KiB into Bytes: **4 × 1024 = 4,096 Bytes [1 mark]**

b) Convert 2.5 MiB into KiB: **2.5 × 1024 = 2,560 KiB [1 mark]**

c) Explain the difference between 1 Kilobyte (KB) and 1 Kibibyte (KiB):

Model Answer: 1 KB uses SI decimal system (10^3 = 1,000 Bytes), whereas 1 KiB uses IEC binary system (2^10 = 1,024 Bytes). [1 mark]

2. Image File Size Calculation: (3 Marks)

A bitmap image has a resolution of **1024 × 768 pixels** and a colour depth of **16 bits per pixel**.

Calculate the file size in **Kibibytes (KiB)**. Show your full working.

Working:

1. Size in bits = 1024 × 768 × 16 = 12,582,912 bits. [1 mark]

2. Size in Bytes = 12,582,912 ÷ 8 = 1,572,864 Bytes. [1 mark]

3. Size in KiB = 1,572,864 ÷ 1024 = 1,536 KiB (or exactly 1.5 MiB). [1 mark]

3. Audio File Size Calculation: (4 Marks)

A studio records a **40-second stereo audio clip** with a sampling rate of **44,100 Hz** and a sample resolution of **16 bits**.

Calculate the size of the uncompressed audio file in **Bytes** and in **Mebibytes (MiB)**.

Working:

1. Bits = Sample Rate × Resolution × Duration × Channels

= 44,100 × 16 × 40 × 2 (stereo) = 56,448,000 bits. [1 mark]

2. Bytes = 56,448,000 ÷ 8 = 7,056,000 Bytes. [1 mark]

3. Size in MiB = 7,056,000 ÷ (1024 × 1024) = 7,056,000 ÷ 1,048,576 ≈ 6.73 MiB. [2 marks]

SECTION 2: DATA COMPRESSION & RUN-LENGTH ENCODING [10 MARKS]

4. Lossy vs Lossless Comparison: (4 Marks)

Complete the comparison table by ticking [✓] or stating True/False:

Statement / Scenario	Lossy	Lossless
Decompressed file is 100% identical to the original uncompressed file.	**No**	**Yes [✓]**
Permanently removes background sounds that human ears cannot perceive.	**Yes [✓]**	**No**
Suitable for executable software program code and medical records.	**No**	**Yes [✓]**
Audio format MP3 and image format JPEG.	**Yes [✓]**	**No**

5. Run-Length Encoding (RLE) Problem Solving: (6 Marks)

a) A sequence of pixel colours is stored as: `W W W W W B B B B G G G G G G W W`

Write the RLE compressed string using (count, colour) pairs: (3 Marks)

Answer: `5W 4B 6G 2W` (or `05 W 04 B 06 G 02 W`). Original = 17 bytes, RLE = 8 bytes. [3 marks]

b) An RLE string is given as: `04 R 03 Y 05 B 02 G`

Decompress the string into its original sequence: (2 Marks)

Answer: `R R R R Y Y Y B B B B B G G` [2 marks]

c) Under what condition does RLE fail to reduce file size? (1 Mark)

Answer: When there are few or no consecutive repeating characters (e.g. `A B C D E F`), each character requires a count byte, doubling the size. [1 mark]

SECTION 3: ERROR DETECTION METHODS [10 MARKS]

6. Parity Block Matrix Error Location: (5 Marks)

A computer transmits data using **Even Parity**. The received parity block grid is shown below:

Byte	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Parity Bit
Byte 1	1	0	1	1	0	0	1	0
Byte 2	0	1	1	0	1	1	0	0
Byte 3	1	1	0	1	0	1	1	1
Byte 4	0	0	1	**0**	1	0	1	1
Parity Byte	0	0	1	**1**	0	0	1	0

a) Identify which byte contains the transmission error: **Byte 4 (has 3 ones with even parity bit 1 → total 4, wait: 0+0+1+0+1+0+1 = 3 ones, parity 1 → Odd parity error in Row 4!) [2 marks]**

b) Identify which bit column flipped: **Bit 4 (column 4 sum = 1+0+1+0+1 = 3 ones, parity 1 → parity error!)**

[2 marks]

c) State the correct value of the corrupted bit: **It should be 1 instead of 0. [1 mark]**

7. Check Digits & ARQ: (5 Marks)

a) Differentiate between a *Transcription Error* and a *Transposition Error*: (2 Marks)

Transcription: Mistyping a single digit (e.g. typing 3 instead of 8).

Transposition: Swapping the position of two adjacent digits (e.g. typing 78 instead of 87). [2 marks]

b) In an ARQ protocol, what happens if the sender does not receive an ACK before the timeout timer expires? (3 Marks)

Model Answer: The timer times out, and the sending computer automatically re-transmits the same packet. This cycle continues until a valid ACK is returned or the retry threshold limit is reached. [3 marks]

SECTION 4: SYMMETRIC VS ASYMMETRIC ENCRYPTION [10 MARKS]

8. Symmetric Encryption Vulnerability: (3 Marks)

Explain why symmetric encryption suffers from the "Key Distribution Problem" and describe how hackers exploit it:

Model Answer: In symmetric encryption, the same key encrypts and decrypts data. To communicate, the sender must send the secret key to the receiver across the network. If an attacker intercepts the transmission of the key, they can decrypt all future confidential messages effortlessly. [3 marks]

9. Asymmetric Public/Private Key Encryption Scenario: (4 Marks)

Zayan wishes to send a confidential, encrypted contract to Nadia over the Internet using asymmetric encryption.

a) Which cryptographic key does Zayan use to encrypt the contract? **Nadia's Public Key [1 mark]**

b) Which key does Nadia use to decrypt and read the contract? **Nadia's Private Key [1 mark]**

c) Can an eavesdropper who intercepts the message decrypt it using Nadia's Public Key? Explain why: (2 Marks)

Answer: No. In asymmetric cryptography, data encrypted with a public key can ONLY be decrypted by the matching private key, which is kept strictly secret on Nadia's device. [2 marks]

10. Digital Signatures & SSL/TLS: (3 Marks)

a) What two security properties does a Digital Signature provide? (2 Marks)

1. Authenticity: Confirms the identity of the true sender.

2. Integrity / Non-repudiation: Proves message was not altered in transit. [2 marks]

b) State the role of a Certificate Authority (CA) in issuing SSL certificates for websites: (1 Mark)

Answer: A trusted third-party organization that verifies the website owner's identity and digitally signs their certificate to prevent spoofing. [1 mark]